INVESTIGATION NOTES

The Imbewu Investigation   Complete Project Documentation

Imbewu Retail | Commercial Analytics | Witle Academy Data Analyst Programme | January 2024 – June 2025

> This document is the complete project investigation record for the Imbewu Retail Western Cape Revenue Investigation capstone. It covers the full four-week investigation: the business context, all four stakeholder messages, every hypothesis formed and tested, all data quality issues discovered, the analytical process, findings, and the reasoning behind each recommendation. It is intended to accompany the investigation.sql, data_dictionary.md, exec_summary.docx, and imbewu_dashboard.pbix as the fifth and final project document.

Contents

1. Project Context & Business Problem

Imbewu Retail is a South African retail chain operating 45 stores across four provinces   Western Cape, Gauteng, KwaZulu-Natal, and Eastern Cape. The stores are split across three formats: Express (smaller, convenience-focused), Market (mid-size), and Mega (large-format).

The dataset covers 18 months of full transactional data from January 2024 through June 2025. It contains 9,164 transactions and 48,641 individual line items, linked to 3,000 loyalty programme customers across Bronze, Silver, and Gold tiers.

The investigation was triggered by a concern from the Head of Sales, Sizwe Khumalo, who observed that Western Cape revenue was declining year-on-year while store visit reports showed foot traffic remaining broadly flat. This apparent contradiction   fewer rands but similar customer counts   was the central puzzle the investigation was designed to solve.

> The challenge of this project was that no specific KPI, no target metric, and no pre-defined question was given. The analyst was expected to structure the investigation themselves, form hypotheses, test them, and present findings to an executive audience with no data background.

2. Stakeholder Messages   All Four Weeks

The project was driven by four stakeholder messages issued one per week. Each message either launched the investigation, added a new request, refined the scope, or changed the output format requirement. All four are recorded here in full as they were received.

Week 1   Sizwe Khumalo, Head of Sales (Monday 09:14)

> Hi team   welcome again to the new joiner. Quick one: I've been looking at the latest revenue report and something's off in Western Cape. Numbers are down compared to last year, but our store visit reports say foot traffic is roughly flat. That doesn't add up. I have an exec readout in three weeks and I need to walk in there knowing what's going on. Can you take this and run with it? I want to know what's happening, why, and what we should do about it. No need to come back with PowerPoint slides   give me one Power BI dashboard I can actually use, plus a short written summary I can forward to the COO. Thanks!

Analytical interpretation: The phrase 'foot traffic is roughly flat' is the key signal. It means customers are still entering the stores   so the problem is not awareness, accessibility, or store closure. Something is changing about what happens inside the store: either customers are buying fewer items, buying cheaper items, or responding to fewer promotions. This shapes the first two hypotheses directly.

Week 2   Naledi Mabaso, Marketing Manager (Friday 15:42)

> Hi! Heard you're the new analyst   welcome. I know you're working on Sizwe's Western Cape thing. While you're in there, can you do me a small favour? We ran the Pap Power Promo (Buy 2 Get 1 Free on Iwisa) in April and the CFO is asking whether it actually worked. I have my own view but I'd love a clean number from someone who isn't already on one side of the argument. Just need a yes/no really, with the numbers behind it. Whenever you have time. Thanks!

Scope management decision: This is a secondary request from a different stakeholder that does not directly address the WC investigation. It was treated as a parallel task   analysed separately and reported in Section 5 of the exec summary. It was not allowed to divert effort away from the core WC hypotheses. The promo analysis was ultimately useful because it provided evidence for the final recommendation in Week 4.

Week 3   Sizwe Khumalo, Head of Sales (Friday 11:08)

> Hi, quick update on what I'm hearing back from the regions. The Western Cape Regional Manager pushed back on me yesterday. She says 'Western Cape is too broad, we have 12 stores and they're not all the same.' Fair point. Can you make sure your analysis goes down to individual store level too? I don't want to say 'WC is down' if it turns out to be 2 or 3 stores dragging the average. We need to know which stores. Same deadline. Thanks!

Analytical interpretation: This message validated Hypothesis 3, which had already been formed independently. The Regional Manager's instinct proved correct   the decline was concentrated in specific stores rather than being uniform across all 12. This message reinforced the importance of store-level granularity and confirmed that presenting a single WC aggregate figure would have been misleading.

Week 4   Vusi Mthembu, Chief Operating Officer (Tuesday 08:02)

> Morning. Sizwe told me you're investigating WC. I'm presenting to the board on Friday week. Forget the long deck   I just want one dashboard I can open on my laptop in front of the board and use to answer questions live. If they ask me 'what's the problem', I open page 1. If they ask 'what's causing it', I open page 2. If they ask 'what should we do', I open page 3. That's the dashboard I want. Make it work.

Dashboard design decision: This message collapsed the original 4-page dashboard plan into a strict 3-page structure. The original plan had separate pages for store performance and category analysis. These were merged into a single 'root cause' page. The Loyalty analysis was also moved to Page 2. A new Page 3 was created specifically for recommendations with supporting evidence. The design principle throughout was: each page answers exactly one board question in the first 10 seconds of viewing.

3. Data Overview & Quality Issues

Before any hypothesis testing began, a full data profiling pass was run across all six tables. The following records what was found and what decisions were made.

Dataset summary

Table	Rows	Description
stores	45	Store metadata: format, province, city, suburb, manager, opened date
customers	3,000	Loyalty programme members with Bronze / Silver / Gold tier
transactions	9,164	Every till receipt issued Jan 2024 – Jun 2025
transaction_items	48,641	Individual product line items inside each transaction
products	48	Product catalogue across 5 categories and multiple sub-categories
promotions	4	Historical marketing campaigns with date ranges and discount percentages

Data quality issues found and decisions made

Issue	Table	Severity	Decision
Mixed province casing: 'western cape' and 'Western Cape' both present	stores	Medium would silently split WC into two groups	Used INITCAP(province) in all SQL queries. Fixed in Power Query via Capitalize Each Word.
~43% of transactions have NULL customer_id	transactions	Medium large segment with no loyalty data	Retained in all revenue calculations. Labelled 'Guest' in loyalty segmentation. Not dropped.
NULL gender on some customer rows	customers	Low	Gender excluded from all analysis. Not relevant to the core investigation.
No direct FK from transactions to promotions	promotions	Medium promo attribution cannot be direct	Inferred by matching transaction dates to promo date ranges and product category.
revenue column in transaction_items loaded as String in Power BI	transaction_items	High breaks all aggregation measures	Fixed in Power Query: changed data type to Decimal Number. Also fixed unit_price_at_sale and discount_applied.

Revenue calculation used throughout

> revenue = (quantity × unit_price_at_sale) - discount_applied

This formula was applied as a calculated column in Power Query and as a DAX measure in Power BI. It reflects the actual net revenue received per line item after any promotional discounts.

4. Analytical Approach   How Hypotheses Were Formed

Before running a single query, four hypotheses were written down based only on the information in Sizwe's message. This is deliberate analytical practice   forming hypotheses before querying prevents confirmation bias and random exploration.

> The instructor tip from the project brief: 'Form hypotheses before you start querying. Write down three or four guesses about what could be happening, even if they feel uninformed. Then design queries to test them. This is how analysts work, not by querying randomly until something interesting appears.'

The four hypotheses were formed as follows:

#	Hypothesis	Reasoning from the brief	Outcome
1	The WC decline is unique and not a national trend	If all provinces are declining, the answer is macro-economic, not WC-specific. Must rule this out first.	Confirmed
2	Basket size is falling, not transaction count	Sizwe said foot traffic is flat. That means transactions are roughly stable. So the revenue loss must come from spend per trip, not number of trips.	Confirmed
3	The decline is concentrated in specific stores, not all 12 WC stores	A regional average can mask individual store performance. Some stores might be growing while a few drag the average down.	Confirmed
4	A specific loyalty segment is driving the spending pullback	If basket size fell, not all customer types fell equally. The customers table and loyalty tier were flagged by the instructor as the most likely source of insight.	Confirmed

All four hypotheses were confirmed by the data. This is unusual   typically one or two are rejected and new ones emerge. In this case the data was consistent and the original framing held throughout the investigation.

5. Hypothesis 1 Is the Decline Unique to Western Cape?

Question being tested

Is WC the only province declining, or is every province down? If national, the cause is external (economy, seasonality). If WC-specific, the cause is local (operational, competitive).

SQL approach

A single CTE query compared H1 2024 vs H1 2025 revenue for each province using conditional aggregation. This avoids self-joins and calculates the YoY change in one pass.

```
WITH revenue_by_period AS (
    SELECT
        INITCAP(s.province) AS province,
        SUM(CASE WHEN t.transaction_date BETWEEN '2024-01-01' AND '2024-06-30'
                 THEN (ti.quantity * ti.unit_price_at_sale) - ti.discount_applied ELSE 0
END) AS rev_h1_2024,
        SUM(CASE WHEN t.transaction_date BETWEEN '2025-01-01' AND '2025-06-30'
                 THEN (ti.quantity * ti.unit_price_at_sale) - ti.discount_applied ELSE 0
END) AS rev_h1_2025
    FROM transaction_items ti
    JOIN transactions t ON ti.transaction_id = t.transaction_id
    JOIN stores s ON t.store_id = s.store_id
    GROUP BY INITCAP(s.province)
)
SELECT province,
    ROUND((rev_h1_2025 - rev_h1_2024) / rev_h1_2024 * 100, 1) AS yoy_change_pct
FROM revenue_by_period ORDER BY yoy_change_pct ASC;
```

Results

Province	H1 2024	H1 2025	YoY %	Status
Western Cape	R386,195	R362,891	-6.0%	Only decline
Eastern Cape	R88,529	R96,153	+8.6%	Growing
Gauteng	R249,259	R273,551	+9.7%	Growing
KwaZulu-Natal	R216,410	R248,124	+14.7%	Growing

Finding

> **CONFIRMED:** Western Cape is the only province declining. All other provinces grew between 8.6% and 14.7% over the same period. The problem is definitively WC-specific. This ruled out macro-economic causes and pointed to a local issue operational, competitive, or customer-driven.

6. Hypothesis 2   Is Basket Size Falling or Transaction Volume?

Question being tested

If foot traffic is flat (as Sizwe stated), then transactions should be roughly stable. The revenue loss must therefore come from smaller baskets   customers spending less per visit   rather than fewer visits.

SQL approach

Transaction count and average basket size were calculated per province per year, allowing a direct comparison of footfall vs spend-per-visit trends.

Results

Province	Year	Transactions	Txn change	Avg basket	Basket change
Western Cape	2024	2,000		R392	
Western Cape	2025	980*	-2.3%	R373	**-4.8%**
Gauteng	2024	1,910		R260	
Gauteng	2025	949*	-0.6%	R289	+11.1%
KwaZulu-Natal	2024	1,536		R296	
KwaZulu-Natal	2025	797*	-0.4%	R313	+5.7%

** H1 only (Jan–Jun). Transaction counts are H1 to H1 comparisons. The proportional change is what matters, not the absolute count.*

Finding

> **CONFIRMED:** Transaction counts across all provinces are broadly flat   this validates Sizwe's statement that foot traffic is unchanged. The WC revenue decline is driven entirely by a falling average basket size: R392 in H1 2024 down to R373 in H1 2025 (-4.8%). Other provinces grew basket size over the same period. Customers are still visiting WC stores   they are leaving with less.

7. Hypothesis 3   Which Specific Stores Are Responsible?

Question being tested

Is the WC decline spread evenly across all 12 stores, or is it concentrated in a small number? If concentrated, the problem is store-specific rather than a WC-wide issue.

This hypothesis was also reinforced by the Week 3 message from Sizwe, passing on feedback from the WC Regional Manager: 'Western Cape is too broad, we have 12 stores and they're not all the same.'

Results   all 12 WC stores ranked by YoY performance

Store	Format	H1 2024	H1 2025	Rand change	YoY %
Mega Bellville	Mega	R73,859	R55,032	-R18,827	**-25.3%**
Market George Central	Market	R26,780	R22,999	-R3,781	-14.1%
Market Claremont	Market	R23,281	R20,044	-R3,237	-13.9%
Mega Paarl Central	Mega	R67,948	R59,794	-R8,154	-12.0%
Express CBD	Express	R8,095	R7,244	-R851	-10.5%
Mega Tygervalley	Mega	R59,662	R55,664	-R3,998	-6.7%
Mega Parow	Mega	R60,417	R61,272	+R855	+1.4%
Express Constantia	Express	R9,178	R10,105	+R927	+10.1%
Market Stellenbosch	Market	R21,590	R23,964	+R2,374	+11.0%
Market Khayelitsha	Market	R22,887	R29,666	+R6,779	+29.6%
Express Sea Point	Express	R6,856	R9,029	+R2,173	+31.7%
Express Mitchells Plain	Express	R5,727	R8,001	+R2,274	+39.7%

Finding

> **CONFIRMED:** The decline is not uniform. It is concentrated in the three Mega stores (Bellville -25.3%, Paarl -12.0%, Tyger valley -6.7%) plus two Market stores (George -14.1%, Claremont -13.9%). Five WC stores grew in H1 2025. The Imbewu brand is not failing across the Western Cape   specific stores in specific locations are underperforming. Bellville Mega alone accounts for R18,827 in lost revenue.

Additionally: the basket size drop at the Mega stores was confirmed. Bellville basket dropped R547 to R463 (-15%). Paarl basket dropped R523 to R464 (-11%). This is not about visit frequency   it is about what customers choose to buy when they visit.

8. Hypothesis 4   Which Customer Segment Is Driving the Pullback?

Question being tested

The customers table contains loyalty tier data (Bronze, Silver, Gold). If the spending decline is concentrated in a specific tier, that identifies the at-risk customer segment and shapes the retention recommendation.

Results   WC revenue by loyalty tier H1 2024 vs H1 2025

Loyalty tier	H1 2024	H1 2025	YoY %	Interpretation
Gold	R89,926	R90,015	+0.1%	Stable top segment held
Bronze	R82,501	R78,719	-4.6%	Mild decline
Guest (no card)	R154,358	R141,484	-8.3%	Significant drop
Silver	R59,410	R52,673	**-11.3%**	Most at-risk segment

Silver tier category breakdown

A second query identified which product categories the Silver tier reduced spend on most sharply in WC:

Category (Silver tier, WC)	H1 2024	H1 2025	YoY %	National YoY %
Health & Beauty	R9,847	R7,945	**-19.3%**	+6.5%
Groceries	R28,504	R23,151	-18.8%	+3.0%
Household	R11,204	R9,447	-15.6%	+0.1%
Apparel	R5,987	R6,215	+3.8%	+14.5%
Electronics	R3,868	R5,915	+52.9%	+6.1%

Finding

> **CONFIRMED:** Gold-tier customers held their spending flat (+0.1%) the most loyal, highest-spending segment did not reduce spend. Silver-tier customers declined by -11.3%, concentrated in Groceries, Household, and Health & Beauty. These categories are declining in WC while growing nationally, which is a WC-specific signal. The pattern is consistent with price-sensitive mid-value customers reducing essential spend   potentially switching to a lower-cost competitor.

9. Stakeholder Side Request   Pap Power Promo Analysis

Requested by Naledi Mabaso (Marketing Manager) during Week 2. The CFO wanted to know whether the Pap Power Promo (Buy 2 Get 1 Free on Iwisa Maize Meal, April 2025) generated real uplift or just moved forward demand that would have occurred anyway.

Approach

April 2025 (promo active) was compared directly to April 2024 (same period, no promo). Both volume and net revenue were measured to assess whether discounts cost more than the volume gains produced.

Results

Metric	April 2024	April 2025	Change	Verdict
Iwisa units sold	146	307	**+110.3%**	Strong uplift
Iwisa net revenue	R5,839	R6,691	+14.6%	Revenue grew
Discount given	R0	R2,466		Cost of promo
Maize Meal category total units	298	453	+52.0%	No cannibalisation
Maize Meal category revenue	R14,536	R14,902	+2.5%	Category also grew

Finding and answer to Naledi

> **VERDICT   YES, the promo worked:** Volume more than doubled (+110%). Net revenue still grew despite the discount (+14.6%). The category as a whole also grew, confirming that Iwisa did not simply cannibalise other Maize Meal brands the promo brought in genuinely incremental volume. The R2,466 in discounts was offset by the revenue and volume gains. This promo model was subsequently recommended as the template for the Silver-tier WC retention campaign in the Week 4 recommendations.

10. Power BI Dashboard   Design Decisions

The dashboard was rebuilt in Week 4 from a 4-page layout to a 3-page layout in response to Vusi Mthembu's (COO) requirement for a board-presentation tool. The design decisions are documented here.

Page structure rationale

Page	Title	Design rationale
1	What's the problem?	Opens with a plain-English insight box. Four KPI cards show the scale. Line chart shows WC diverging from national trend. Bar chart shows WC as the only negative province. Province summary table gives the full numbers.
2	What's causing it?	Diverging store bar chart is the hero visual immediately shows Bellville and Paarl as outliers. Format breakdown and category charts provide the 'why'. Loyalty tier bars show the Silver segment story. Key findings boxes give plain-English summaries for each underperforming store.
3	What should we do?	Three action cards (Immediate / Short-term / Investigate) with colour-coded urgency. Pap Power Promo table provides evidence that the recommended campaign mechanic is proven. Bright spots table shows WC is not fundamentally broken it shows the growth playbook.

Key design principles applied

- **One board question per page:** Each page is designed so the exec can answer the board's question in the first 10 seconds without reading body text.
- **Insight box first:** Every page opens with a plain-English sentence stating the answer before any chart is seen. The board member reads the conclusion, then looks at the evidence.
- **Red for decline, green for growth:** Consistent throughout. No mixed signals.
- **No chart titles using default Power BI labels:** All axis labels and chart titles were renamed to plain English.
- **Page navigation buttons:** Added so Vusi can move between pages during live board questions without touching the Power BI tab bar.

11. Final Findings Summary

Western Cape revenue declined by 6.0% in H1 2025 versus H1 2024. This is a WC-specific problem   every other province grew. The root cause is a falling average basket size (-4.8%), not fewer store visits. The decline is concentrated in three Mega stores (Bellville, Paarl, Tygervalley) and driven by Silver-tier loyalty customers reducing spend on Groceries and Household items. Five WC stores grew in the same period, confirming the brand is not the problem.

#	Finding	Evidence
1	WC revenue down -6.0% while all other provinces grew	Province YoY comparison: EC +8.6%, Gauteng +9.7%, KZN +14.7%
2	Basket size fell (-4.8%), transactions flat (-2.3%)	Avg basket R392 → R373. Transaction count broadly unchanged YoY.
3	3 Mega stores account for 132% of the gross WC revenue loss	Bellville -R18,827, Paarl -R8,154, Tygervalley -R3,998 vs 5 growing stores offsetting partially.
4	Silver-tier customers are the most at-risk segment (-11.3%)	Gold held flat at +0.1%. Silver down -11.3%, concentrated in Groceries, Household, H&Beauty.
5	Pap Power Promo model is proven (+110% volume, +14.6% revenue)	April 2025 vs April 2024 Iwisa comparison. Incremental no cannibalisation detected.

12. Recommendations with Evidence

Immediate action (0–4 weeks)

Audit Bellville Mega, Paarl Mega, and Tygervalley Mega: These three stores account for R30,819 in lost revenue   132% of the total WC gap. The remaining stores' growth partially offsets this but cannot compensate fully. A store-level operational audit should focus on Groceries and Household shelf availability, product pricing versus local competitors, and proximity to any new retail entrants. Bellville is the highest priority: -R18,827 alone, and Groceries accounted for 65% of Bellville's total decline (lost R12,402 in that category alone). The basket dropped from R547 to R463 (-15%)   customers are actively buying less per trip.

Short-term action (1–3 months)

Retain Silver-tier WC loyalty customers: Silver-tier members are visiting 10.3% less often and spending less per trip. Gold tier is stable. The priority is to re-engage Silver members before further churn occurs. Run a targeted Groceries or Household offer exclusively for Silver-tier WC members. The Pap Power Promo (Buy 2 Get 1) is the proven template   it delivered +110% volume and +14.6% net revenue in April 2025. However, the 33.3% discount structure was aggressive. Recommend a 20% discount for the Silver-tier retention campaign to achieve similar volume uplift while protecting margin more effectively.

Ongoing investigation

Investigate competitive threats near Bellville and Paarl: The Bellville Groceries decline pattern   high volume, fast decline, concentrated in everyday staples   is the signature of a new competitor opening nearby and capturing regular shopping trips. Tygervalley's flat visits with a shrinking basket is also consistent with price comparison behaviour. Recommended actions: (1) check for new grocery or discount retail store openings near Bellville and Paarl within the last 12 months; (2) survey WC Silver-tier members who have not transacted in 90+ days; (3) review Bellville's pricing on the top 10 Groceries SKUs versus local competitors.

13. What Was Learned

Analytical skills developed

- **Hypothesis-first investigation:** Forming four testable hypotheses before opening a SQL editor prevented random exploration and kept the investigation focused. All four were confirmed, validating the approach.
- **Conditional aggregation for YoY comparisons:** Using SUM(CASE WHEN transaction_date BETWEEN...) inside a single CTE produced clean YoY comparisons without self-joins, which would have been significantly more complex.
- **The difference between a foot traffic problem and a basket size problem:** These require completely different responses. A foot traffic problem is solved by marketing and awareness. A basket size problem is solved by in-store experience, pricing, promotions, and product range. Identifying the correct root cause shaped every recommendation.
- **NULL handling in loyalty analysis:** ~43% of transactions had no customer_id. Dropping these would have removed nearly half the revenue from loyalty analysis. Using COALESCE to label them as 'Guest' retained the full picture.
- **Data type errors in Power BI:** The revenue column loaded as String type, breaking all aggregation measures with a SUM cannot work with values of type String error. This was fixed in Power Query by changing the data type to Decimal Number before building any visuals.

Communication and stakeholder management

- **Scope management across four stakeholders:** Four different people added requests over four weeks. Only the requests that served the core investigation question were given primary effort. Naledi's promo request was treated as a parallel task and folded into the recommendations, not treated as a distraction.
- **The Regional Manager's challenge was right:** Sizwe's Week 3 message passing on the WC Regional Manager's feedback 'WC is too broad' turned out to be exactly correct. Presenting a single WC aggregate would have missed the store-level story entirely.
- **Executive dashboards require ruthless simplicity:** Vusi's Week 4 message forced a reduction from 4 pages to 3. This made the dashboard better. The discipline of one question per page, one insight box per page, and no more than four visuals per page created a presentation tool that a COO could actually use in front of a board.
- **Every chart should answer a question, not display data:** The difference between a data display and an insight is the plain-English title and insight box. A chart titled 'Revenue by Province 2024 vs 2025' is data. A chart titled 'H1 Revenue Growth by Province WC is the only decline' is an insight.